

# What Needs an Authority

Mechanical Detection, Chartered Resolution, and the Exact Price of Sole Ownership

Paper 6 of the Harbor program: the coordination functions of a multi-agent harbor,  
sorted into those an algorithm discharges and those a charter must pay for

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## Abstract

Multi-agent governance documents hand out authorities freely: a *lookout* that scans incoming commitments for conflicts, a *sole owner* for the roadmap, the test suite, the release tag. This paper asks which of those coordination functions genuinely require a designated authority, and prices the ones that do. Two results. First, conflict *detection* needs no authority at all — inside a designed commitment fragment  $\mathcal{L}_c$  (definite Horn rules with  $\perp$  heads, ground deontic obligation/prohibition rules over scopes and intervals, exclusive interval claims, difference constraints), detection is a polynomial, witness-producing algorithm,  $O(H + T \log T + C \log C + V \cdot E)$ , validated against a brute-force oracle on 3000 random policy sets with zero disagreements; and composed with the supervisory-control theorem of Paper 2 it is not even a role, but a guard: acceptance is a mediated write, so in-fragment conflict-freedom is *regimentable* — the runtime refuses the commit. One expressive step outside the fragment (disjunctive obligations under discharge choice), conflict-freedom is NP-complete, and *there* an authority becomes necessary: not to notice the conflict but to choose the discharge. Second, sole *ownership* of a role is not a principle but an inequality: a sole specialist beats a pooled swarm exactly when the skill premium clears the Erlang-C boundary  $g_A(\rho, c)$  — the whitepaper’s proposed threshold is falsified in both directions, with the two thresholds crossing at  $\rho = (3 - \sqrt{5})/2$  — and a sole owner who can die and be succeeded is viable at *any* skill premium only while the death-rate-times-succession-time product stays below  $D^* = \eta K / (1 - \eta K)$ : past that line, pooling wins even against an infinitely skilled specialist. The slogan the two theorems earn: an authority is needed exactly where the algorithm ends, and where it is needed, its price is a computable number, not a vibe.

**Express lane.** *Conflict detection inside the designed commitment fragment is polynomial and witness-producing (no judge consumed, and via Paper 2 the check is enforceable pre-commit by the runtime); one disjunctive obligation outside the fragment makes conflict-freedom NP-complete, which is where a chartered resolver must choose; and sole ownership of a role pays iff  $\mu_s/\mu_g \geq g_A(\rho, c)$  — exact Erlang-C form in the box of §6 — with sole-owner viability capped at  $\xi/\eta \leq D^*$  absent skill limits.* Experts: read the two boxes (§3, §6), the succession box (§7), and the inventory table (§8); the rest is scaffolding.

## 1 The scene: the harbor’s authority inventory

A harbor of coding agents runs on commitments: standing prohibitions (*never write to prod*), conditional obligations (*if you deploy, you must update the runbook within the window*), exclusive claims (*this file is mine from tick 5 to tick 20*), deadlines chained to deadlines. The governance document responds with authorities. A *lookout* is chartered to scan every proposed commitment against the accepted set and reject contradictions. *Sole-responsibility roles* — a roadmap owner,

a test-suite curator, a release avatar — each hold exclusive write capability over one invariant, on the theory that a single deeply-skilled owner beats a committee.

Both assignments feel natural, and both smuggle in an unexamined premise: that the function *requires a designated authority* — a mind with standing, judgment, and a name — rather than a subroutine and a price. The operations lead’s two questions are therefore exact. *Does the lookout need judgment, or is contradiction-scanning mechanical? And when does the sole owner actually beat the pool — and what happens when the owner dies mid-quarter?* This paper answers both with theorems that were run before they were believed: the first by a 3000-instance oracle sweep with zero disagreements [internal, `b4_deontic_fragment.py`], the second by a sweep that *falsified the whitepaper’s own proposed threshold in both directions* before replacing it [internal, `b8_specialization.py`].

### The idea in one breath.

*An authority is needed exactly where the algorithm ends: inside a designed commitment fragment, conflict detection is a polynomial witness-producing decision the runtime can enforce pre-commit — no judge; one expressive step outside, conflict-freedom is NP-complete and a chartered resolver must choose; and sole ownership of a role is justified not by principle but by an exact Erlang-C threshold on skill, payable only while the owner’s death-rate-times-succession-time stays below  $D^*$ .*

**Intuition: the working port.** Watch a real port for a day (Figure 1). Whether two vessels are on a collision course is *geometry* — constant bearing, decreasing range — computed identically by every watch officer; nobody’s authority is consulted. Who gives way is next decided by a *rulebook* (the collision regulations) — still mechanical, still nobody’s judgment. Only where the rulebook ends — a three-vessel crossing the rules never anticipated — does harbor traffic control exercise chartered discretion. And the port’s one genuinely personal authority, the licensed pilot with twenty years on these shoals, is hired against a pool of tug crews only when the water is hard enough to pay the premium — and the port licenses a second pilot, because a port whose only pilot can fall ill has priced its channel’s availability at one human body. The structural mapping: collision geometry  $\rightarrow$  in-fragment conflict detection (mechanical, witness-producing); the rulebook’s edge  $\rightarrow$  the fragment’s NP-complete frontier (where chartered resolution begins); the pilot’s premium and the second license  $\rightarrow$  the  $g_A(\rho, c)$  boundary and the succession price  $D^*$ . The analogy maps relations, so it earns a prediction: making the rulebook *richer* should eventually make rule-following itself intractable and reintroduce discretion — and that is exactly what the NP-completeness theorem says happens one disjunction past the fragment.

The predictable misread, preempted: *mechanical detection does not mean mechanical governance*. The checker certifies the norm *system*, never the norms — a bad policy consistently held is consistent — and detection without a resolution charter just produces alarms. The theorems do not abolish authority; they *relocate* it, from noticing to choosing, and then hand the remaining authority its bill.

## 2 Part I. The lookout is a subroutine: the fragment $\mathcal{L}_c$

Cheap detection is a *language-design* choice, not a discovery about deontic logic in general. The commitment language  $\mathcal{L}_c$  buys it by construction. A policy set in  $\mathcal{L}_c$  contains:

- **facts** (ground atoms) and **definite Horn rules**  $a_1 \wedge \dots \wedge a_k \rightarrow b$ , where the head  $b$  may be the contradiction symbol  $\perp$ ;
- **deontic rules**  $\text{body} \rightarrow M(\text{act}, s, [t_1, t_2])$  with  $M \in \{O, F\}$ : when the (Horn) body holds, action  $\text{act}$  is obliged or forbidden on scope atoms  $s$  over the ground time interval  $[t_1, t_2]$ ;
- **exclusive interval claims** (*resource  $x$  is exclusively mine on  $[t_1, t_2]$* );

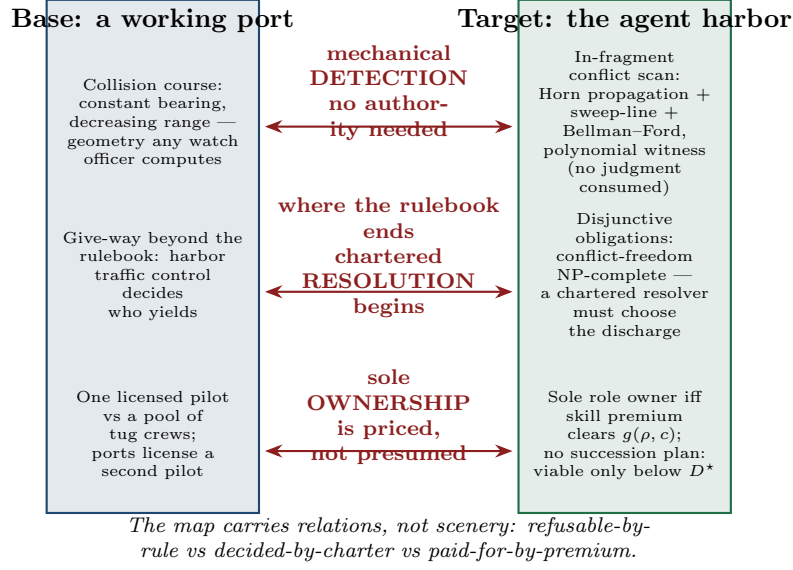


Figure 1: The relation map. Base domain: a working port — collision-course detection is geometry, give-way beyond the rulebook is harbor traffic control’s chartered call, and the sole licensed pilot is paid a premium against a pool of tug crews, with a second license mandated. Target domain: the agent harbor — in-fragment conflict scanning is Horn propagation + sweep-line + Bellman-Ford with a polynomial witness; disjunctive obligations make conflict-freedom NP-complete, so a chartered resolver chooses the discharge; and a sole role owner is justified iff the skill premium clears  $g_A(\rho, c)$ , with no-succession viability capped at  $D^*$ . The correspondence arrows carry the theorem content: refusable-by-rule vs decided-by-charter vs paid-for-by-premium.

- **difference constraints**  $x_j - x_i \leq c$  over deadline variables.

A *conflict* is any of four things: derivable  $\perp$ ; a fired O/F pair on the same action with overlapping scope-and-interval; two exclusive claims overlapping on the same resource; or a negative cycle in the difference-constraint graph (an unsatisfiable deadline system). This is deliberately a *ground* fragment — no nesting of deontic operators, no negation-as-failure, monotone bodies only — and §9 says explicitly which classical deontic puzzles are thereby left outside the door.

### 3 The detection theorem and its frontier

**Theorem 1a (membership: detection is polynomial and witness-producing).** For a policy set in  $\mathcal{L}_c$  with Horn program size  $H$ ,  $T$  fired deontic tokens,  $C$  exclusive claims, and difference-constraint graph  $(V, E)$ , conflict detection is decidable in  $O(H + T \log T + C \log C + V \cdot E)$ , with a polynomial witness per conflict: Dowling–Gallier unit propagation computes the least Horn model in  $O(H)$  — which by monotonicity equals the intersection of all models, i.e. exactly the statuses forced in *every* world [verified framework, Dowling–Gallier 1984]; a keyed sweep-line over interval endpoints finds every O/F clash and claim overlap in  $O(T \log T + C \log C)$ ; Bellman–Ford returns a negative cycle iff the deadline system is infeasible, in  $O(V \cdot E)$ . The witnesses are, respectively: a derivation chain, the two fired rules with their overlap interval, the two claims, and the negative cycle.

**Theorem 1b (frontier: one step outside is NP-complete).** Extend  $\mathcal{L}_c$  with dis-

junctive obligations  $O(l_1 \vee \dots \vee l_k)$  under discharge-choice semantics (the agent picks which disjunct to discharge). Conflict-freeness of a policy set becomes NP-complete: membership by the selection certificate; hardness from 3-SAT — each variable  $x$  becomes a pair of deontic rules  $\text{sel}_x^+ \rightarrow O(\text{assert}_x)$  and  $\text{sel}_x^- \rightarrow F(\text{assert}_x)$  on *identical* scope and interval, each clause becomes a disjunctive obligation over its selector literals, and a conflict-free discharge selection exists iff the formula is satisfiable [verified: reduction restated here in full; checked both directions on 16/16 instances, internal, `b4_deontic_fragment.py`].

**Verification.** 3000 random in-fragment policy sets against a brute-force oracle (all  $2^5$  Horn models intersected; pairwise overlap checks; exhaustive integer search over the complete potential box  $[-\Sigma|c|, 0]^V$  for the difference system): 885 conflicts found — 121 derivable  $\perp$ , 195 O/F clashes, 484 claim overlaps, 213 negative temporal cycles — of which 149 are reachable *only* through Horn propagation; **0 disagreements** with the oracle, and least model = intersection-of-all-models on all 3000 [internal, `b4_deontic_fragment.py`, seed 20260816]. The 3-SAT reduction was verified in both directions on 8 satisfiable and 8 unsatisfiable random instances ( $n=4$ ,  $m=7-28$ ), 16/16, with the satisfying assignment extracted from each conflict-free selection. Mutation: deleting Horn propagation (checking only directly stated statuses) misses 100 of the 885 sweep conflicts — propagation is load-bearing at scale, not a formality [internal].

**Numbers by hand.** The mutant’s canonical miss is a four-line policy a reviewer can check aloud: fact `agent_deployed`; Horn rule `agent_deployed`  $\rightarrow$  `prod_access`; deontic rule `prod_access`  $\rightarrow$   $O(\text{write\_prod}, s, [0, 10])$ ; standing  $F(\text{write\_prod}, s, [5, 15])$ . Propagation forces `prod_access`, the obligation fires, and the sweep-line reports the O/F overlap on  $[0, 10] \cap [5, 15] = [5, 10]$  — while a pairwise scanner that never propagates sees no directly stated clash and stays silent [internal, the printed witness]. For the frontier, run the reduction on the smallest unsatisfiable formula  $(x) \wedge (\neg x)$ : two disjunctive obligations force both  $\text{sel}_x^+$  and  $\text{sel}_x^-$ , firing  $O(\text{assert}_x)$  and  $F(\text{assert}_x)$  on identical scope and interval — every selection conflicts, matching unsatisfiability [verified: two lines]. *Now you try:* facts `{audit_open}`; rules `audit_open`  $\rightarrow$  `vault_access` and `vault_access`  $\rightarrow$   $O(\text{read\_ledger}, s, [2, 6])$ ; standing  $F(\text{read\_ledger}, s, [4, 9])$ . Conflict? (Yes: propagation forces `vault_access`, the obligation fires, and the overlap is  $[4, 6]$  — a two-hop indirect conflict, exactly the species the mutant misses.)

## 4 Composition with Paper 2: the lookout is not even a role

Paper 2 proved that a safety policy over an agent’s event alphabet is enforceable-by-prevention (*regimentable*) iff it is Ramadge–Wonham controllable — no uncontrollable event carries a legal history out of the policy — and its design rule was: uncontrollable events in a policy’s *condition* are free; only uncontrollable events in its *prohibition* are fatal. Now compose. In the harbor, *acceptance of a commitment is a mediated write*: the accepted set lives in the single-writer store, and the acceptance event is a member of  $\Sigma_c$ , the controllable alphabet. The policy “*never accept a proposal whose union with the accepted set conflicts in-fragment*” therefore gates a controllable event on a condition that Theorem 1a makes a total, polynomial-time-computable predicate of committed state. The policy is controllable, hence regimentable: the conflict never enters the accepted set, because the daemon refuses the write, with the theorem’s witness attached to the refusal [verified: composition of Theorem 1a with Paper 2’s boxed theorem; both components machine-checked separately, internal, `b4_deontic_fragment.py` + `b3_controllability.py`].

This is the paper’s first answer, and it is sharper than “the lookout doesn’t need judgment.” Inside the fragment the lookout is not an authority at all — it is a *guard clause* at the mediation boundary, as impersonal as a type check, producing a four-line witness instead of a verdict. What the charter must still supply is everything the guard cannot: *which* of two conflicting proposals yields (priority, negotiation, waiver), *whether* a standing norm should be amended,

and — the moment anyone wants disjunctive obligations — *which discharge to choose*, because Theorem 1b says no proposal-time check scales there unless  $P=NP$ . The frontier is thus the exact line where authority re-enters: the schema designer reads Theorem 1 as a price list — keep the language in  $\mathcal{L}_c$  and detection is a regimentable subroutine; buy the expressive convenience of disjunction and you have hired a judge, with a queue.

## 5 Part II. The sole owner: an inequality, not a principle

Part I located the one place inside conflict management where a mind is genuinely required: not to notice a contradiction but to choose a discharge among disjuncts the fragment cannot rank on its own. Part II asks the harbor’s other standing authority question, about a different kind of decision entirely — not *which* candidate resolution to pick, but *whether one designated person should be doing the picking at all* — and it gets the same kind of answer: not a principle to invoke but a number to compute. The whitepaper’s sole-responsibility roles (roadmap owner, test-suite curator, release avatar) trade pooled capacity for accountability and skill. Model the trade honestly. Requests for the invariant-critical function arrive Poisson at rate  $\lambda$ ; a sole specialist serves them as an  $M/M/1$  queue at rate  $\mu_s$ ; the pooled alternative is an  $M/M/c$  queue of generalists at rate  $\mu_g$  each, utilization  $\rho = \lambda/(c\mu_g)$ ; the skill premium is  $r = \mu_s/\mu_g$ . Sole ownership also carries an *accountability value*  $A$  (the worth of single-writer attribution per unit time of demand) against a waiting cost  $w$  per request-hour. The whitepaper proposed the threshold  $\tilde{g} = 1 + (c - 1)\rho/(1 - \rho)$  and marked it *status: proposed*. The sweep ran before the belief formed — and falsified  $\tilde{g}$  in both directions.

## 6 The specialization boundary

**Theorem 2 (queueing specialization boundary).** With  $C(c, \rho)$  the Erlang-C delay probability, sole ownership weakly dominates the pool in net cost iff the skill premium clears

$$\frac{\mu_s}{\mu_g} \geq g_A(\rho, c) = c\rho + \frac{1}{1 + \frac{C(c, \rho)}{c(1 - \rho)} + \frac{A\mu_g}{w\lambda}},$$

reducing at  $A = 0$  to  $g(\rho, c) = c\rho + \frac{c(1 - \rho)}{c(1 - \rho) + C(c, \rho)}$ , with  $g(\rho, 1) = 1$  (a specialist against one generalist needs no premium),  $g \rightarrow c$  as  $\rho \rightarrow 1$  (the boundary *saturates* at the capacity ratio, never diverges), and the closed form  $g(\rho, 2) = 1 + 2\rho - \rho^2$  [verified: derivation below]. The proposed threshold  $\tilde{g} = 1 + (c - 1)\rho/(1 - \rho)$  is **falsified in both directions**: at  $c=2, \rho=0.1, r=1.15$  it certifies a specialist who loses ( $W_{\text{solo}} = 1.0526 > W_{\text{pool}} = 1.0101$ ; simulation  $z = -15.7$ ), at  $c=2, \rho=0.5, r=1.9$  it rejects a specialist who wins ( $1.111 < 1.333$ ;  $z = +32$ ); the two thresholds cross at  $\rho = (3 - \sqrt{5})/2 \approx 0.382$ , and where  $g$  saturates  $\tilde{g}$  diverges ( $\rho=0.9, c=8$ :  $g = 7.73$  vs  $\tilde{g} = 64$ ) [internal, `b8.specialization.py`, seed 20260816].

**Numbers by hand.** The theorem is two lines of algebra on textbook formulas, which is the right way to first believe it. Sole ownership pays iff  $w\lambda(W_{\text{solo}} - W_{\text{pool}}) \leq A$ , i.e.  $1/(\mu_s - \lambda) \leq W_{\text{pool}} + A/(w\lambda)$ ; invert and divide by  $\mu_g$ , using  $\lambda/\mu_g = c\rho$  and  $\mu_g W_{\text{pool}} = 1 + C(c, \rho)/(c(1 - \rho))$ , and  $g_A$  falls out [verified]. For  $c = 2$  the Erlang-C value is  $C(2, \rho) = 2\rho^2/(1 + \rho)$ , so  $c(1 - \rho)/(c(1 - \rho) + C) = (1 - \rho^2)/((1 - \rho^2) + \rho^2) = 1 - \rho^2$  and  $g(\rho, 2) = 2\rho + 1 - \rho^2$  [verified: three lines]. Now check the falsification instances by hand at  $\mu_g = 1$ . At  $\rho = 0.1$ :  $g = 1 + 0.2 - 0.01 = 1.19$  while  $\tilde{g} = 1 + 0.1/0.9 = 1.111$ , so  $r = 1.15$  sits between them — the whitepaper certifies, the truth rejects; indeed  $W_{\text{solo}} = 1/(1.15 - 0.2) = 1.0526$  against  $W_{\text{pool}} = 1 + 0.0182/1.8 = 1.0101$  [verified].

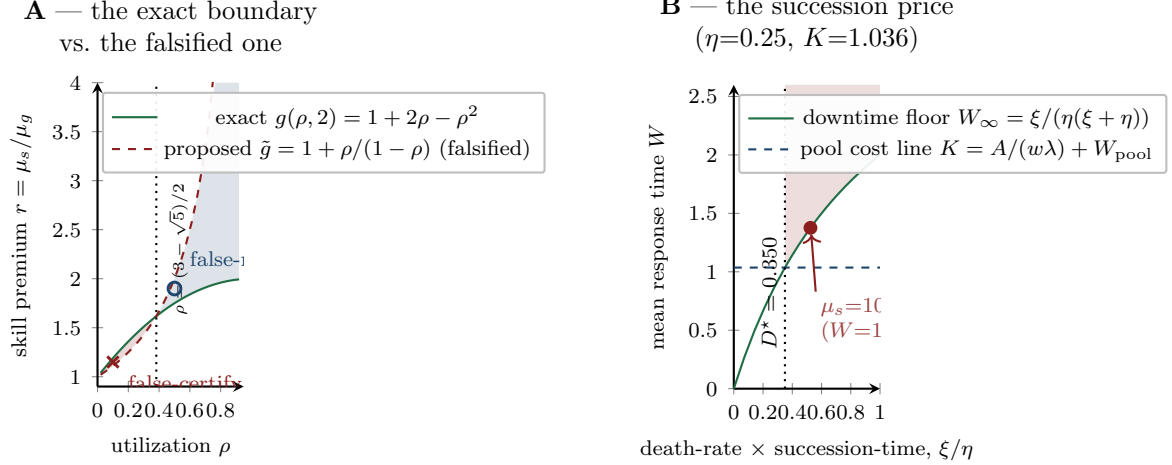


Figure 2: The regime diagram. Panel A: the exact boundary  $g(\rho, 2) = 1 + 2\rho - \rho^2$  (solid) against the falsified proposal  $\tilde{g}$  (dashed); the shaded lenses are the two error regimes — false-certify below the crossing at  $\rho = (3 - \sqrt{5})/2$ , false-reject above it — with the two refuting instances marked ( $r=1.15$  at  $\rho=0.1$ ;  $r=1.9$  at  $\rho=0.5$ ). Panel B: the succession price at the reference instance ( $\eta=0.25$ ,  $K=1.036$ ): the downtime floor  $W_\infty = \xi/(\eta(\xi + \eta))$  crosses the pool cost line at  $D^* = 0.350$ ; to its right pooling wins at every skill premium — the marked point is a  $\mu_s = 10^8$  specialist still losing at  $\xi/\eta = 1.5D^*$  ( $W = 1.376 > K$ ). Both panels regenerate from `b8_specialization.py` [internal, seed 20260816].

At  $\rho = 0.5$ :  $g = 1.75$ ,  $\tilde{g} = 2$ , and  $r = 1.9$  again sits between — rejected by the whitepaper, but  $W_{\text{solo}} = 1/0.9 = 1.111 < W_{\text{pool}} = 4/3$  [verified]. Setting  $1 + 2\rho - \rho^2 = 1 + \rho/(1 - \rho)$  gives  $\rho^2 - 3\rho + 1 = 0$ : the thresholds cross at  $\rho = (3 - \sqrt{5})/2$ , which is why the falsification runs in *both* directions — Figure 2, Panel A, shades the two error lenses [verified]. *Now you try*: at  $c=2$ ,  $\rho=0.3$ , does  $r = 1.5$  justify a sole owner? ( $g(0.3, 2) = 1 + 0.6 - 0.09 = 1.51$  — no, by a hair;  $\tilde{g} = 1.43$  would have said yes. The flattering threshold flatters exactly below the crossing.)

The cross-validation stack behind the closed forms: Erlang-C means against discrete-event simulation,  $|z| \leq 1.8$  across the grid; the 60-instance random sweep over  $(\lambda, \mu_s, \mu_g, c, A, w, \xi, \eta)$  gave 55 decisive instances and 0 sign violations of the boundary [internal, `b8_specialization.py`].

## 7 The succession price: what one mortal body costs

The boundary above assumes the specialist never disappears. Sole owners disappear: sessions die, containers are reaped, providers churn. Model the corner honestly: the specialist *dies* at rate  $\xi$  and is *succeeded* after an  $\text{Exp}(\eta)$  spin-up (context transfer, warm-up, credential rebinding), work resuming where it stopped, requests queueing throughout — an  $M/M/1$  queue with breakdowns in the Mitrany–Avi-Itzhak lineage.

**Theorem 3 (price of the succession rule).** With effective rate  $\mu_{\text{eff}} = \mu_s \eta / (\xi + \eta)$ , the sole owner’s mean response time is exactly

$$W_{\text{bd}} = \frac{1 + \mu_s \xi / (\xi + \eta)^2}{\mu_{\text{eff}} - \lambda} = \frac{1}{\mu_{\text{eff}} - \lambda} + \frac{\mu_{\text{eff}}}{\mu_{\text{eff}} - \lambda} W_\infty, \quad W_\infty = \frac{\xi}{\eta(\xi + \eta)},$$

decreasing in  $\mu_s$  with infimum  $W_\infty$ : *no amount of skill buys back residual downtime*. Consequently, with  $K = A/(w\lambda) + W_{\text{pool}}$  the pool’s cost line, pooling dominates at *every* skill

premium iff

$$\frac{\xi}{\eta} > D^* = \frac{\eta K}{1 - \eta K},$$

and  $D^*$  is the price of the succession rule: a sole role without a succession plan (small  $\eta$ ) is viable only while its death-rate-times-succession-time product stays below  $D^*$  [verified:  $W_\infty(\xi^*) = K$  at  $\xi^* = D^*\eta$  is one line of algebra; closed form checked against matrix-geometric solution to  $2 \times 10^{-13}$ , truncated CTMC to  $10^{-14}$ , Gillespie simulation  $|z| \leq 0.8$ , internal, `b8_specialization.py`].

**Numbers by hand.** Reference instance:  $\lambda=1$ ,  $c=4$ ,  $\mu_g=1.2$  (so  $\rho = 0.208$ ),  $A=0.2$ ,  $w=1$ ,  $\eta=0.25$ . Erlang-C gives  $C(4, 0.208) = 0.0110$ , so  $W_{\text{pool}} = 1/1.2 + 0.0110/(4 \cdot 1.2 \cdot 0.792) = 0.8362$  and  $K = 0.2 + 0.8362 = 1.0362$  [verified: four-function-calculator arithmetic on the boxed formulas]; then  $\eta K = 0.2591$  and  $D^* = 0.2591/0.7409 = 0.350$  [verified]. Just past the line, at  $\xi/\eta = 1.5 D^*$ , an absurdly skilled specialist ( $\mu_s = 10^8$ ) still loses:  $W = 1.376 > K = 1.036$  — the floor  $W_\infty$ , not the skill, is binding [internal]. The identity in the box also explains the program’s recorded wrong turn: the naive decomposition  $W_{\text{bd}} = 1/(\mu_{\text{eff}} - \lambda) + W_\infty$  is refuted (it predicts 4.17 where the truth is 8.17 at the canary instance  $\mu_s=5$ ,  $\xi=0.5$ ), because downtime also *builds queue*: the residual is amplified by  $\mu_{\text{eff}}/(\mu_{\text{eff}} - \lambda) = 2.5$  there, and  $1.5 + 2.5 \times 2.667 = 8.17$  checks by hand [verified arithmetic; refutation internal]. Panel B of Figure 2 draws exactly this geometry: the floor  $W_\infty$  crossing the pool cost line at  $D^*$ . *Now you try*: at the reference instance, is a sole owner with a same-day succession plan ( $\eta = 2$ ;  $K$  is unchanged at 1.0362, so  $\eta K = 2.07 > 1$ ) ever capped? (No:  $\eta K \geq 1$  makes  $D^* = \infty$  — a fast enough succession plan removes the viability ceiling entirely, leaving only the Theorem-2 premium test. That is what a succession plan *buys*.)

The implementation hazard is also priced. A cost model that forgets breakdowns entirely (plain  $M/M/1$ ) is caught by the high- $\xi$  canary: at  $\xi/\eta = 2 \gg D^*$ , the mutant says SPECIALIST while simulation says the pool wins decisively ( $\Delta C = -7.06 \pm 0.16$ ); at low  $\xi$  the mutant is indistinguishable — so the canary, not routine testing, is what makes the omission visible [internal, `b8_specialization.py`].

The predictable misread, preempted:  $D^*$  does not say specialists are bad, and Theorem 2 does not say pooling is the default truth. Together they say sole ownership is a *purchase*: below the boundary the premium does not cover the congestion cost and the honest ledger entry is “we pay latency  $w\lambda(W_{\text{solo}} - W_{\text{pool}}) - A$  per unit time for accountability”; above  $D^*$  the purchase is unavailable at any price until the succession plan (larger  $\eta$ ) is bought first. Every quantity in both tests —  $\lambda$ ,  $\mu_s$ ,  $\mu_g$ ,  $\xi$ ,  $\eta$  — is daemon-metered, so the harbor can re-grade its roles from telemetry rather than from taste.

## 8 The authority inventory, priced

The two parts assemble into the answer to the title question:

| Coordination function                         | Authority needed?      | What it costs                                                             |
|-----------------------------------------------|------------------------|---------------------------------------------------------------------------|
| In-fragment conflict detection                | none (algorithm)       | $O(H + T \log T + C \log C + V \cdot E)$<br>per batch, witness included   |
| In-fragment conflict enforcement              | none (runtime guard)   | mediated acceptance write; regimentable via Paper 2                       |
| Conflict-freedom with disjunctive obligations | chartered resolver     | NP-complete check, or a judge who chooses the discharge                   |
| Priority, waiver, norm amendment              | chartered, irreducibly | outside every checker; the charter’s residual job                         |
| Sole ownership of a role                      | conditional            | justified iff $r \geq g_A(\rho, c)$ ; otherwise a priced latency purchase |
| Sole ownership without succession             | conditional, capped    | viable only while $\xi/\eta \leq D^* = \eta K / (1 - \eta K)$             |

The first two rows delete an authority the governance document thought it needed; the third and fourth rows locate the authority that is genuinely irreducible; the last two rows keep the authority but hand it an invoice. That is the paper’s claim in institutional terms: a well-designed harbor spends judgment *only* where theorems certify judgment is required, and meters what the remaining judgment costs.

## 9 Related work

**Imported.** Linear-time Horn satisfiability and unit propagation are Dowling–Gallier [1]; we import the algorithm and the least-model property unchanged, and our fragment is *designed* so that it applies. NP-completeness machinery is Cook and Karp [2, 3]; the reduction in Theorem 1b is standard 3-SAT gadgetry, and the contribution is only the *location* of the frontier — discharge-choice disjunction — in a commitment language a scheduler would actually want. The regimentation composition imports Ramadge–Wonham supervisory control [4] via Paper 2. On the queueing side, the Erlang-C delay formula is a century old [5], the economies of pooling in many-server queues are classical folklore sharpened by Halfin–Whitt [6], and queues with server breakdowns descend from Mitrany–Avi-Itzhak [7]; Theorem 3’s  $W_{bd}$  is in that lineage. **Positioned against deontic logic.** The fragment is ground and operator-nesting-free: conflicts are defined operationally (O vs F on overlapping scope and interval), not as theoremhood in standard deontic logic [8], and the classical paradoxes — Ross’s paradox, Chisholm’s contrary-to-duty puzzle [9] — simply cannot be *stated* in  $\mathcal{L}_c$ : that is a designed retreat, declared rather than hidden, and it is what the tractability is paid for with. **New, honestly.** No component algorithm or queueing formula above is ours. The contributions are: the fragment/frontier pair as an exact price list for commitment-language design; the composition making in-fragment conflict-freedom regimentable rather than merely checkable; the falsification and replacement of the whitepaper’s specialization threshold by the exact  $g_A(\rho, c)$  with its  $c=2$  closed form and crossing point; and the succession criterion  $D^*$  stating, in daemon-metered quantities, when no skill premium can rescue a sole role. An August-2026 survey found no prior statement of the authority question in this detection/resolution/ownership decomposition; a final lit-sweep before submission is owed — “not found” is not “proven nonexistent.”

## 10 Honest boundary

**What these theorems do NOT say.** (i) Theorem 1’s completeness is relative to least-model semantics: monotone bodies only — negation-as-failure, unification, and symbolic interval endpoints coupled to difference variables all leave the fragment, and nothing here

is claimed about them. (ii) The batch complexity bound is what is proven; the incremental amortized form (watched literals, interval trees) is a constants engineering claim, stated in the portfolio, not certified here. (iii) The checker certifies the norm *system*, never the norms: a harbor whose policies are consistently harmful passes every scan — norm content is chartered authority’s irreducible job, row four of the inventory. (iv) The regimentation composition inherits Paper 2’s deployment obligation wholesale: acceptance must *actually* be completely mediated ( $\Sigma_c$  honest), which a red-team, not this proof, discharges. (v) Theorem 2 compares *mean* costs only — no tail percentiles, no SLA quantiles — under FCFS and exponential everything;  $A$  and  $w$  are policy prices the theorem constrains but does not pick. (vi) Theorem 3’s breakdown model is death-at-all-times with preemptive resume (the death/heartbeat model); active-only breakdowns would let sufficient skill escape the  $W_\infty$  floor, so the model choice is load-bearing and declared. (vii) The wrong turns are on record and priced: the naive  $W_{bd}$  decomposition is refuted (4.17 vs 8.17), and the whitepaper’s  $\tilde{g}$  was falsified by this program’s own sweep — the correction, not the original, is what ships. (viii) The inventory covers detection, resolution, and ownership; other authority functions (charter amendment, dispute appeal, norm authorship) are chartered by construction and no theorem here bears on them.

*Reproducibility.* Every [internal] number regenerates from two self-contained scripts at seed 20260816: `skills/harbor-results/scripts/b4_deontic_fragment.py` (3000-instance oracle sweep, conflict census, least-model check, 16/16 reduction verification, Horn-propagation mutation) and `skills/harbor-results/scripts/b8_specialization.py` (falsification instances, Erlang-C vs DES, matrix-geometric / CTMC / Gillespie cross-validation of  $W_{bd}$ , 60-instance sign sweep,  $D^*$  instance, breakdown-forgetting mutation canary). Every [verified] number is a closed form or short derivation restated in this paper’s boxes and checkable by hand or from the cited literature. The two figures regenerate from `docs/harbor-research/figures/src/paper6_figures.py`.

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